

Department of Mathematics

Course Content

Course Name: Numerical

Course Code: CSE 0541-2205

Teacher Name: Md. Shah Najmus Shakib

Course Credit: 03

CIE Marks: 90

SEE marks: 60

Semester End Exam (SEE): 03 hours

Class:	17 weeks (2 class per week) = 34 Hours
Holyday/ Leave:	3 weeks
Exam/ Result:	06 weeks
Total:	34 class per semester

Course Learning Outcome (CLO): After successful completion of the course students will be able to -

CLO1	Student will acquire proficiency in fundamental concepts and apply it to solve different problems.
CL02	Develop advanced analytical and problem-solving skills essential for tackling complex computational challenges.
CLO3	Demonstrate mastery several mathematical proofs and apply the method to solve problems.
CLO4	Apply theoretical knowledge to effectively solve practical problems.

Assessment Pattern

Continuous Internal Evaluation (CIE 90 marks)

Blooms Category	Test (Out of 45)	Assignments (15)	Quiz (15)	Co-curricular Activities (15)
Remember	05		5	Attendance 15
Understand	05			
Apply	10			
Analysis	8	7	10	
Evaluate	7	8		
Create	10			

Semester End Examination (SEE 60)

Blooms Category	Test (Out of 60)
Remember	10
Understand	10
Apply	10
Analysis	10
Evaluate	10
Create	10

SL.	Content of Courses	Hrs	CLO's
1	Solving non-linear equations such as Bi-Section, False Position, Newton-Raphson, Fixed Point Iteration, and Bairstow's Method. Solving system of linear equations solutions including Triangular systems and back substitution.	10	CLO1, CLO2
2	LU-factorization, Pivoting, Interpolation and approximation techniques like Newton's forward and backward interpolation, Divided differences formula, Lagrange polynomial.	8	CLO1, CLO4
3	Richardson's extrapolation, and numerical differentiation techniques	8	CLO1, CLO3
4	Trapezoidal rule, Simpson's rule (1/3 rule and 3/8 rule), and Romberg's integration, Euler's method, Modified Euler's method	8	CLO2, CLO4

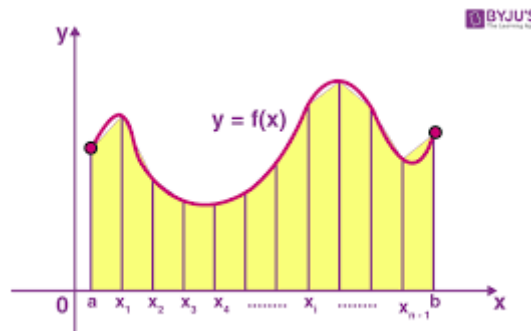
Course Plan Specific Content, CLOs, Teaching Learning and Assessment Strategy mapped with CLOs.

Week No.	Task Heading	Topics	Teaching-Learning Strategy	Assessment Strategy	Corresponding CLO's
1	Numerical Integration	Trapezoidal rule	Lecture, Discussion	Presentation, Quiz	CLO2, CLO4
2	Numerical Integration	Simpson's rule 1/3 rule	Lecture, Discussion	Quiz, Written Assignment, Oral Presentation	CLO2, CLO4
3	Numerical differentiation	Simpson's rule 3/8 rule	Lecture, Discussion	Oral Presentation, Group Assignment	CLO2, CLO4
4	Numerical differentiation	Richardson's extrapolation	Lecture, Discussion	Oral Presentation	CLO1, CLO3
5	Solution of Non-linear Equations	Bisection method	Discussion, Oral Presentation	Written Assignment	CLO1, CLO2
6	Solution of Non-linear Equations	False Position method	Discussion, Oral Presentation	Written Assignment	CLO1, CLO2
7	Solution of Non-linear Equations	Newton-Raphson method	Oral Presentation	Oral Presentation	CLO1, CLO2
8	Solution of Non-linear Equations	Fixed Point Iteration	Group Work	Group Assignment	CLO1, CLO2
9	Solution of system of Linear equations	Triangular systems and back substitution, Forward and Backward Substitution	Group Work	Quiz, Written Assignment	CLO1, CLO2
10	Solution of system of Linear equations	Gauss elimination method	Lecture, Discussion	Oral Presentation, Quiz	CLO1, CLO2
11	Solution of system of Linear equations	LU-factorization	Discussion, Oral Presentation	Group Assignment, Quiz	CLO1, CLO4

12	Solution of system of Linear equations	Pivoting	Oral Presentation	Presentation, Written Assignment	CLO1, CLO2
13	Interpolation and Approximation	Newton's forward and backward interpolation	Oral Presentation	Quiz, Presentation	CLO1, CLO2,
14	Interpolation and Approximation	Newton's forward and backward interpolation	Group Work	Written Assignment, Oral Presentation	CLO1, CLO2, CLO3
15	Interpolation and Approximation	Divided differences formula	Discussion, Oral Presentation	Group Assignment, Presentation	CLO1, CLO2
16	Interpolation and Approximation	Lagrangian interpolation	Discussion, Oral Presentation	Quiz, Group Assignment	CLO1, CLO2
17	Interpolation and Approximation	Lagrangian interpolation	Oral Presentation	Written Assignment, Quiz	CLO1, CLO2

Algorithm for Trapezoidal Rule:

1. Define a function $f(x)$
2. Set a = lower limit and b = upper limit
3. Calculate step size $h = \frac{b-a}{n}$, where n is even.
4. Calculate $x_i = a + ih$ and $y_i = f(x_i)$, $i = 0, 1, 2, \dots, n$
5. Integration value $I = \frac{h}{2} [y_0 + y_n + 2(y_1 + y_2 + y_3 + \dots + y_{n-1})]$



Question: Compute the integral $\int_0^1 \frac{1}{\sqrt{1+x^2}} dx$ by using **Trapezoidal rule**.

Solution: Here $a = 0$ and $b = 1$, we shall divide the interval into 10 equal parts.

Hence $h = \frac{1-0}{10} = 0.1$ Now we find the values of $y = \frac{1}{\sqrt{1+x^2}}$ for each point of subdivision

in the following table.

x	y
$x_0 = 0$	$y_0 = 1$
$x_1 = 0.1$	$y_1 = 0.995037$
$x_2 = 0.2$	$y_2 = 0.980580$
$x_3 = 0.3$	$y_3 = 0.957826$
$x_4 = 0.4$	$y_4 = 0.928476$
$x_5 = 0.5$	$y_5 = 0.894427$
$x_6 = 0.6$	$y_6 = 0.857493$
$x_7 = 0.7$	$y_7 = 0.819232$
$x_8 = 0.8$	$y_8 = 0.780869$

$x_9 = 0.9$	$y_9 = 0.743294$
$x_{10} = 1.0$	$y_{10} = 0.707107$

By Trapezoidal rule, we have

$$\begin{aligned} \int_0^1 \frac{1}{\sqrt{1+x^2}} dx &= \frac{h}{2} [(y_0 + y_{10}) + 2(y_1 + y_2 + y_3 + y_4 + y_5 + y_6 + y_7 + y_8 + y_9)] \\ &= \frac{0.1}{2} [1 + 0.7071 + 2(0.9950 + 0.9806 + 0.9578 + 0.9285 + 0.8944 + 0.8575 + 0.8192 + 0.7809 + 0.7433)] \\ &= 0.8811 \end{aligned}$$

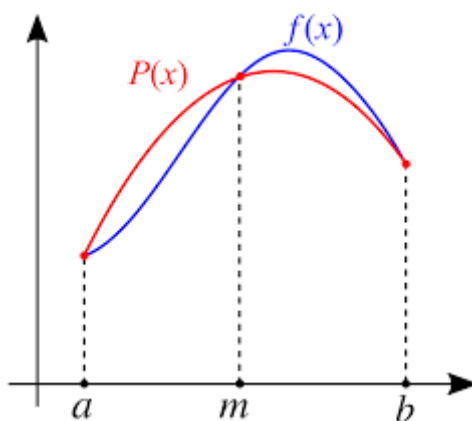
2nd Week

Topic: Simpson's 1/3 rule

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Algorithm for Simpson's 1/3 rule:

1. Define a function $f(x)$
2. Set a = lower limit and b = upper limit
3. Calculate step size $h = \frac{b-a}{n}$, where n is the multiple of 6.
4. Calculate $x_i = a + ih$ and $y_i = f(x_i)$, $i = 0, 1, 2, \dots, n$
5. Integration value $I = \frac{h}{3} [(y_0 + y_n) + 4(y_1 + y_3 + \dots + y_{n-1}) + 2(y_2 + y_4 + \dots + y_{n-2})]$



Question: Compute the integral $\int_0^6 \frac{1}{(1+x)^2} dx$ by using Simpson's $\frac{1}{3}$ rule.

Solution: Here $a = 0$ and $b = 6$, we shall divide the interval into 6 equal parts.

Hence $h = \frac{6-0}{6} = 1$ Now we find the values of $y = \frac{1}{(1+x)^2}$ for each point of subdivision in

the following table.

x	y
$x_0 = 0$	$y_0 = 1$
$x_1 = 1$	$y_1 = 0.25000$
$x_2 = 2$	$y_2 = 0.11111$
$x_3 = 3$	$y_3 = 0.06250$
$x_4 = 4$	$y_4 = 0.04000$
$x_5 = 5$	$y_5 = 0.02778$
$x_6 = 6$	$y_6 = 0.02041$

By Simpson's $\frac{1}{3}$ rule, we have

$$\begin{aligned}
 \int_0^6 \frac{1}{(1+x)^2} dx &= \frac{h}{3} [(y_0 + y_6) + 4(y_1 + y_3 + y_5) + 2(y_2 + y_4)] \\
 &= \frac{1}{3} [(1 + 0.02041) + 4(0.25000 + 0.06250 + 0.02778) + 2(0.11111 + 0.04000)] \\
 &= 0.89458
 \end{aligned}$$

Week: 3

Topic: Simpson's 3/8 rule

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Algorithm for Simpson's 3/8 rule:

1. Define a function $f(x)$
2. Set a = lower limit and b = upper limit
3. Calculate step size $h = \frac{b-a}{n}$, where n is the multiple of 6.
4. Calculate $x_i = a + ih$ and $y_i = f(x_i)$, $i = 0, 1, 2, \dots, n$
5. Integration value

$$I = \frac{3h}{8} [(y_0 + y_n) + 3(y_1 + y_2 + y_4 + \dots + y_{n-1}) + 2(y_3 + y_6 + \dots + y_{n-3})]$$

Question: Compute the integral $\int_0^1 \frac{1}{1+x} dx$ by using **Simpson's $\frac{3}{8}$ rule**.

Solution: Here $a = 0$ and $b = 1$, we shall divide the interval into 6 equal parts.

Hence $h = \frac{1-0}{6} = \frac{1}{6}$ Now we find the values of $y = \frac{1}{1+x}$ for each point of subdivision in the following table.

x	y
$x_0 = 0$	$y_0 = 1$
$x_1 = \frac{1}{6}$	$y_1 = 0.857143$
$x_2 = \frac{2}{6}$	$y_2 = 0.750000$
$x_3 = \frac{3}{6}$	$y_3 = 0.666667$
$x_4 = \frac{4}{6}$	$y_4 = 0.600000$
$x_5 = \frac{5}{6}$	$y_5 = 0.545455$
$x_6 = 1$	$y_6 = 0.500000$

By Simpson's $\frac{3}{8}$ rule, we have

$$\begin{aligned}
 \int_0^1 \frac{1}{1+x} dx &= \frac{3h}{8} [(y_0 + y_6) + 3(y_1 + y_2 + y_4 + y_5) + 2y_3] \\
 &= \frac{3}{48} [(1 + 0.5) + 3(0.857143 + 0.75 + 0.60 + 0.545455) + 2 \times 0.666667] \\
 &= 0.6931955
 \end{aligned}$$

Numerical Differentiation and Integration

Question: Compute the integral $\int_4^{5.2} \ln x \, dx$ by using Trapezoidal, Simpson's $\frac{1}{3}$ and

Simpson's $\frac{3}{8}$ rule.

Solution: Here $a = 4$ and $b = 5.2$, we shall divide the interval into $n=6$ equals parts. $h = \frac{b-a}{n}$

Hence $h = \frac{5.2-4}{6} = 0.2$ Now we find the values of $y = \ln x$ for each point of subdivision in the following table.

x	y
$x_0 = 4$	$y_0 = 1.38629$
$x_1 = 4.2$	$y_1 = 1.43508$
$x_2 = 4.4$	$y_2 = 1.48160$

$x_3 = 4.6$	$y_3 = 1.52606$
$x_4 = 4.8$	$y_4 = 1.56862$
$x_5 = 5.0$	$y_5 = 1.60944$
$x_6 = 5.2$	$y_6 = 1.64866$

By Trapezoidal rule, we have

$$\begin{aligned}
 \int_4^{5.2} \ln x \, dx &= \frac{h}{2} \left[(y_0 + y_6) + 2(y_1 + y_2 + y_3 + y_4 + y_5) \right] \\
 &= \frac{0.2}{3} \left[(1.38629 + 1.64866) + 2(1.43508 + 1.48160 + 1.52606 + 1.56862 + 1.60944) \right] \\
 &= 1.82766
 \end{aligned}$$

.....

By Simpson's $\frac{1}{3}$ rule, we have

$$\begin{aligned}
 \int_4^{5.2} \ln x \, dx &= \frac{h}{3} \left[(y_0 + y_6) + 4(y_1 + y_3 + y_5) + 2(y_2 + y_4) \right] \\
 &= \frac{0.2}{3} \left[(1.38629 + 1.64866) + 4(1.43508 + 1.52606 + 1.60944) + 2(1.48160 + 1.56862) \right] \\
 &= 1.82785
 \end{aligned}$$

By Simpson's $\frac{3}{8}$ rule, we have

$$\begin{aligned}
 \int_4^{5.2} \ln x \, dx &= \frac{3h}{8} \left[(y_0 + y_6) + 3(y_1 + y_2 + y_4 + y_5) + 2y_3 \right] \\
 &= \frac{3 \times 0.2}{8} \left[(1.38629 + 1.64866) + 3(1.43508 + 1.48160 + 1.56862 + 1.60944) + 2 \times 1.52606 \right] \\
 &= 1.82785
 \end{aligned}$$

Exercise: Compute the integral the following integral by using Trapezoidal, Simpson's $\frac{1}{3}$

and Simpson's $\frac{3}{8}$ rule.

$$(i) \int_0^1 \frac{1}{1+x^2} \, dx \quad (ii) \int_0^{\frac{\pi}{2}} \sin x \, dx \quad (iii) \int_{0.2}^{1.4} (\sin x - \ln x + e^x) \, dx$$

Solution: Here $a = 0.2$ and $b = 1.4$, we shall divide the interval into 6 equal parts.

Hence $h = \frac{1.4 - 0.2}{6} = 0.2$ Now we find the values of $y = \ln x$ for each point of subdivision in the following table.

x	$y = \sin x - \ln x + e^x$
$x_0 = 0.2$	$y_0 = 3.029510001$
$x_1 = 0.4$	$y_1 = 2.797533772$
$x_2 = 0.6$	$y_2 = 2.897586898$
$x_3 = 0.8$	$y_3 = 3.166040571$
$x_4 = 1.0$	$y_4 = 3.559752813$
$x_5 = 1.2$	$y_5 = 4.069834452$
$x_6 = 1.4$	$y_6 = 4.70417746$

Week: 4

Topic: Richardson extrapolation

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1. $f(x) = x \exp(x)$, find $f'(2)$ with $h=0.2$ using Richardson extrapolation method

Solution:

Richardson Extrapolation method

Equation is $f(x) = x \exp(x)$.

$$h = 0.2$$

$$f'(x) = \frac{f(x+h) - f(x-h)}{2h} \quad (\text{CDD})$$

$$f'(2) = \frac{f(2+0.2) - f(2-0.2)}{2 \cdot 0.2}$$

$$f'(2) = \frac{f(2.2) - f(1.8)}{0.4}$$

$$f'(2) = \frac{19.855 - 10.8894}{0.4}$$

$$f'(2) = 22.4142$$

$$h = 0.1$$

$$f'(x) = \frac{f(x+h) - f(x-h)}{2h}$$

$$f'(2) = \frac{f(2+0.1) - f(2-0.1)}{2 \cdot 0.1}$$

$$f'(2) = \frac{f(2.1) - f(1.9)}{0.2}$$

$$f'(2) = \frac{17.149 - 12.7032}{0.2}$$

$$f'(2) = 22.2288$$

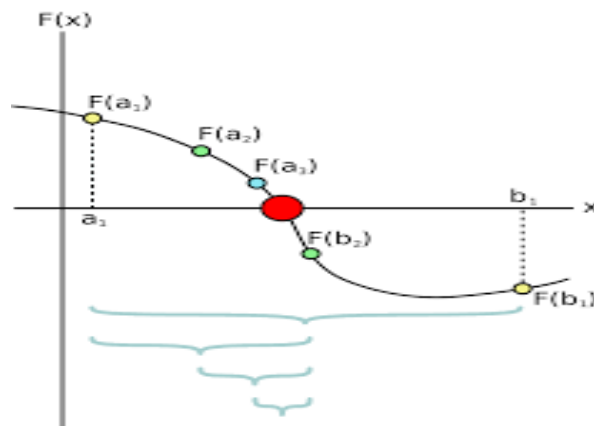
$$TV = (AV)_{\frac{h}{2}} + \frac{(AV)_{\frac{h}{2}} - (AV)_h}{3}$$

$$= 22.2288 + \frac{22.2288 - 22.4142}{3}$$

$$= 22.167$$

Algorithm of Bisection Method:

1. Define a function $f(x)$
2. Find two points a and b so that $f(a) < 0$ and $f(b) > 0$.
3. Define a new point $c = \frac{(a+b)}{2}$
4. If $f(c) < 0$, then set $a = c$ otherwise set $b = c$
5. Continuing the process till the desired place of decimals is not obtained.



Question: Determine the root of $x^2 - 3 = 0$ for $x \in [1, 2]$ by using **Bisection Method**.

Solution: Given: $x^2 - 3 = 0$ Let $f(x) = x^2 - 3$

Now, find the value of $f(x)$ between $a = 1$ and $b = 2$.

$$f(1) = 1^2 - 3 = -2 < 0 \quad \text{and} \quad f(2) = 2^2 - 3 = 1 > 0$$

The given function is continuous, and the root lies in the interval $[1, 2]$.

The iterations for the given functions are:

Iterations	a	b	$c = \frac{a+b}{2}$	$f(c)$
1	1.0	2.0	1.5	-0.7500
2	1.5	2.0	1.75	0.06250
3	1.5	1.75	1.625	-0.3594
4	1.625	1.75	1.6875	-0.1523

5	1.6875	1.75	1.71875	-0.04590
6	1.71875	1.75	1.734375	0.008057
7	1.71875	1.734375	1.7265625	-0.01898
8	1.7265625	1.734375	1.7304688	-0.005478
9	1.7304688	1.734375	1.7324219	0.001286
10	1.7304688	1.7324219	1.7314453	-0.002097
11	1.7314453	1.7324219	1.7319336	-0.000406
12	1.7319336	1.7324219	1.7321777	0.0004397
13	1.7319336	1.7321777	1.7320557	0.0000168

Since $|x_{13} - x_{12}| = 1.22 \times 10^{-4} < 1 \times 10^{-3}$. Hence, 1.7320557 is the approximated solution.

Question: Find the root of the equation $e^{-x}(3.2\sin x - 0.5\cos x) = 0$ for $x \in [0, 0.5]$ by using

Bisection Method.

Solution: Let $f(x) = e^{-x}(3.2\sin x - 0.5\cos x)$.

Now, find the value of $f(x)$ at $a = 0$ and $b = 0.5$.

$$f(0) = e^{-0}(3.2\sin 0 - 0.5\cos 0) = -0.5 < 0$$

$$\text{and } f(0.5) = e^{-0.5}(3.2\sin 0.5 - 0.5\cos 0.5) = 0.664 > 0$$

The given function is continuous, and the root lies in the interval $[0, 0.5]$.

The iterations for the given functions are:

Iterations	a	b	$c = \frac{a+b}{2}$	$f(c)$
1	0	0.5	0.25	0.239276
2	0	0.25	0.125	-0.085725
3	0.125	0.25	0.1875	0.087259
4	0.125	0.1875	0.15625	0.003472
5	0.125	0.15625	0.140625	-0.040440
6	0.140625	0.15625	0.1484375	-0.018314
7	0.1484375	0.15625	0.15234375	-0.007378
8	0.15234375	0.15625	0.15429688	-0.001943
9	0.15429688	0.15625	0.15527344	0.000767
10	0.15429688	0.15527344	0.15478516	-0.000587

11	0.15478516	0.15527344	0.15502930	0.000090
12	0.15478516	0.15502930	0.15490723	-0.000248
13	0.15490723	0.15502930	0.15496826	-0.000079

Since $|x_{13} - x_{12}| = 6.1 \times 10^{-5} < 1 \times 10^{-4}$. Hence, 0.154968 is the approximated solution.

Exercise: Find the values of the following functions by using Bisection method.

(i) $x^3 - x - 1 = 0$ (ii) $2x - 3\sin x - 5 = 0$ (iii) $x^3 - 4x - 5 = 0$ (iv) $xe^x - 1 = 0$

Week: 6

Topic: False Position Method/Regular Falsii Method

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Algorithm of False Position Method:

1. Define a function $f(x)$
2. Find two points a and b so that $f(a) < 0$ and $f(b) > 0$.
3. Define a new point $c = \frac{a f(b) - b f(a)}{f(b) - f(a)}$
4. If $f(c) < 0$, then set $a = c$ otherwise set $b = c$
5. Continuing the process till the desired place of decimals is not obtained.

Question: Determine the root of $2x^3 - 2x - 5 = 0$ for $x \in [1.2, 1.8]$ using **False Position**

Method. Tol = 10^{-6}

Solution: Given: $2x^3 - 2x - 5 = 0$ Let $f(x) = 2x^3 - 2x - 5$

Now, find the value of $f(x)$ between $a = 1.2$ and $b = 1.8$.

$$f(1.2) = 2(1.2)^3 - 2 \times 1.2 - 5 = -3.944 < 0 \quad \text{and} \quad f(1.8) = 2(1.8)^3 - 2 \times 1.8 - 5 = 3.064 > 0$$

The given function is continuous, and the root lies in the interval $[1.2, 1.8]$.

The iterations for the given functions are:

Iterations	a	b	c $= \frac{a f(b) - b f(a)}{f(b) - f(a)}$	$f(c)$

1	1.2	1.8	1.53767123	-0.80390181
2	1.53767123	1.80000000	1.59219345	-0.11171146
3	1.59219345	1.80000000	1.59950342	-0.01463191
4	1.59950342	1.80000000	1.60045633	-0.00190145
5	1.60045633	1.80000000	1.60058008	-0.00024684
6	1.60058008	1.80000000	1.60059615	-0.00003204
7	1.60059615	1.80000000	1.60059823	0.0000041588
8	1.60059823	1.80000000	1.60059850	-0.000000539

Since $|x_8 - x_7| = 2.7 \times 10^{-7} < 1 \times 10^{-6}$. Hence, 1.60059850 is the approximated solution.

Exercise: Find the values of the following functions by using False Position method.

(i) $x^3 - 2x - 5 = 0$ (ii) $3x - \cos x - 1 = 0$ (iii) $x^2 - \ln x - 12 = 0$

Week: 7

Topic: Newton-Raphson Method

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Algorithm of Newton-Raphson Method:

1. Define a function $f(x)$
2. Find points a and b such that $a < b$ and $f(a) \times f(b) < 0$
3. Take the interval $[a, b]$ and find next value $x_0 = \frac{a+b}{2}$
4. Find the derivative of the function $f'(x)$
5. Find $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$
6. If $f(x_n) = 0$ then x_n is an exact root, otherwise $x_n = x_{n+1}$
7. Repeat the step 5 and 6 until $f(x_i) = 0$, $i = 1, 2, 3, \dots$ or $|f(x_i)| \leq \text{Accuracy}$.

Question: Find the positive root of $x = \cos x$ for $x \in [0, 1]$ using Newton-Raphson method.

Let $f(x) = x - \cos x \therefore f'(x) = 1 + \sin x$

Here $f(0) = -1 < 0$ and $f(1) = 0.4596769 > 0$

The given function is continuous, and the root lies in the interval $[0, 1]$.

Here, $x_0 = \frac{0+1}{2} = 0.5$.

By Newton-Raphson method, we have $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$

$$\Rightarrow x_{n+1} = x_n - \frac{x_n - \cos x_n}{1 + \sin x_n} = \frac{x_n \sin x_n + \cos x_n}{1 + \sin x_n}$$

Taking $x_0 = 0.5$, we get

$$\text{For, } n=0, x_1 = \frac{x_0 \sin x_0 + \cos x_0}{1 + \sin x_0} = \frac{0.5 \sin 0.5 + \cos 0.5}{1 + \sin 0.5} = 0.755222417$$

$$\text{For, } n=1, x_2 = 0.739141666$$

$$x_3 = 0.739085134$$

$$x_4 = 0.739085133$$

Since $|x_4 - x_3| = 4 \times 10^{-9} < 1 \times 10^{-8}$. Hence the required root is 0.739085133

Home works:

1. $x^3 - 2x^2 - 4 = 0$, $x \in [2, 3]$, correct up to 5 decimal places
2. $x^4 + x^2 - 80 = 0$, $x \in [2, 3]$, correct up to 3 decimal places
3. $e^{-x} = 10x$, $x \in [0, 0.2]$, correct up to 8 decimal places
4. $x = \sqrt[3]{48}$, $x \in [3, 2.4]$, correct up to 8 decimal places

Exercise: Find the values of the following functions by using Newton-Raphson method.

- (i) $2x^3 - 3x - 6 = 0$; $[1, 2]$ (ii) $x e^x - \cos x = 0$; $[0, 1]$
 (iii) $3x - \cos x - 1 = 0$; $[0, 1]$ (iv) $10x - \sin x - 6 = 0$; $[0, 1]$
 (i) $f'(x) = 6x^2 - 3$ (ii) $f'(x) = x e^x + e^x + \sin x$ (iii) $f'(x) = 3 + \sin x$ (iv) $f'(x) = 10 - \cos x$

Week: 8

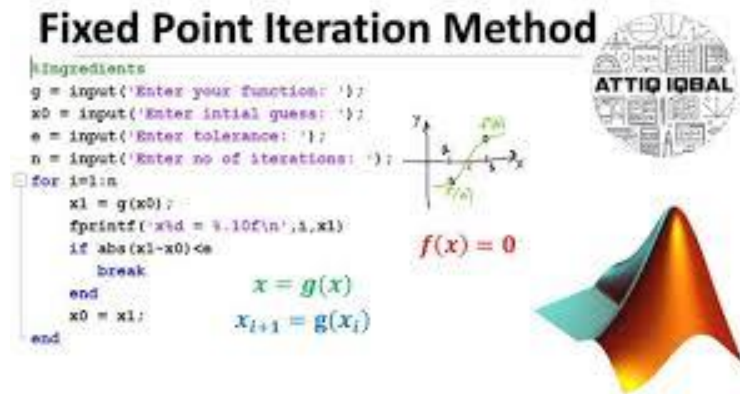
Topic: Fixed Point Iteration Method

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Algorithm of Fixed Point Iteration Method:

1. Define a function $f(x)$
2. Find points a and b such that $a < b$ and $f(a) \times f(b) < 0$
3. Take the interval $[a, b]$ and find next value $x_0 = \frac{a+b}{2}$
4. Define function $g(x)$ which is obtained from $f(x) = 0$ such that $x = g(x)$
5. Guess a value x_0 such that $|g'(x_0)| < 1$
6. Find $x_{n+1} = g(x_n)$
7. If $f(x_n) = 0$ then x_n is an exact root, otherwise set $x_n = x_{n+1}$

8. Repeat the step 4 and 5 until $f(x_i) = 0$, $i = 1, 2, 3, \dots$ or $|x_{i+1} - x_i| \leq \text{Accuracy}$.



Question: Find the root of $x^2 - x - 1 = 0$ for $x \in [1, 2]$ by using **Fixed Point Iteration** method.

Solution: Given $x^2 - x - 1 = 0$. Let $f(x) = x^2 - x - 1$.

Here $f(1) = -1 < 0$ and $f(2) = 1$. A root lies between 1 and 2.

The given equation can be written as $x = \sqrt{x_n + 1} = g(x_n)$

Here $g'(x) = \frac{1}{2\sqrt{x+1}}$ and $|g'(x)| = \left| \frac{1}{2\sqrt{x+1}} \right|$ for all $x \in [1, 2]$

Here, $x_0 = \frac{1+2}{2} = 1.5$

$|g'(1.5)| = \left| \frac{1}{2\sqrt{1.5+1}} \right| = 0.3162 < 1$

By Iteration method, we have $x_{n+1} = g(x_n)$

For, $n=0$, $x_1 = g(x_0) = \sqrt{x_0 + 1}$

$$x_1 = \sqrt{x_0 + 1} = 1.58113883$$

$$x_2 = 1.606592304$$

$$x_3 = 1.614494442$$

$$x_4 = 1.616939839$$

$$x_5 = 1.617695842$$

$$x_6 = 1.617929492$$

$$x_7 = 1.618001697$$

$$x_8 = 1.618024010$$

Since $|x_8 - x_7| = 2.23 \times 10^{-5} < 1 \times 10^{-4}$. Hence the desired root is 1.61802401

Home works:

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Exercise: Find the values of the following functions by using Fixed Point Iteration method.

$$(i)x^3 - 2x^2 - 4 = 0; [2,3] \quad (ii) \sin^2 x - x^2 + 1 = 0; [1,2] \quad (iii)x^3 + x^2 - 1 = 0; [0,1] \quad (iv)2x - \cos x - 3 = 0; [1,2]$$

$$(i)g(x) = \sqrt[3]{2x^2 + 4} \quad (ii)g(x) = \sqrt{1 + (\sin(x))^2} \quad (iii)g(x) = \frac{1}{\sqrt{1+x}} \quad (iv)g(x) = \frac{\cos x + 3}{2}$$

Week: 9

Topic: Triangular systems and back substitution, Forward and Backward Substitution

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Triangular Systems and Back Substitution

A triangular system refers to a system of linear equations where the coefficient matrix is either upper or lower triangular. In an upper triangular system, all the elements below the diagonal are zero, while in a lower triangular system, all elements above the diagonal are zero. Solving triangular systems is efficient because it allows for straightforward methods like forward and backward substitution.

Forward Substitution is used when solving a system with a lower triangular matrix. Here's how it works:

1. Start with the first equation, which involves only one variable.
2. Solve for that variable.
3. Substitute the value into the next equation, which will have two variables.
4. Continue substituting solved values into subsequent equations until all variables are determined.

This method is called "forward" because we solve for variables from top to bottom, progressing through the equations in sequence.

Backward Substitution, on the other hand, is used for upper triangular systems. Here's the process:

1. Begin with the last equation, which contains only one variable.
2. Solve for this variable.
3. Substitute the value into the second-to-last equation, which will have two variables.
4. Continue substituting solved values into previous equations until all variables are determined.

Backward substitution is called so because we solve for variables starting from the bottom and working upwards, utilizing already-known values for higher variables.

Both substitution methods are efficient and provide a systematic approach to solving triangular systems. They are commonly used in numerical methods and algorithms like Gaussian elimination, especially when dealing with large systems of equations where direct methods may be computationally expensive.

In summary, triangular systems and the associated forward and backward substitution methods simplify the solution of linear systems by breaking down the equations in a step-by-step manner.

Week: 10

Topic: Gauss elimination method

Page: 20

1. Solve Equations $2x+5y=21, x+2y=8$ using Gauss-Jordan Elimination method

Solution:

Total Equations are 2

$$2x + 5y = 21 \rightarrow (1)$$

$$x + 2y = 8 \rightarrow (2)$$

Converting given equations into matrix form

$$\left[\begin{array}{cc|c} 2 & 5 & 21 \\ 1 & 2 & 8 \end{array} \right]$$

$$R_1 \leftarrow R_1 \div 2$$

$$= \left[\begin{array}{cc|c} 1 & \frac{5}{2} & \frac{21}{2} \\ 1 & 2 & 8 \end{array} \right]$$

$$R_2 \leftarrow R_2 - R_1$$

$$= \left[\begin{array}{cc|c} 1 & \frac{5}{2} & \frac{21}{2} \\ 0 & -\frac{1}{2} & -\frac{5}{2} \end{array} \right]$$

$$R_2 \leftarrow R_2 \times -2$$

$$= \left[\begin{array}{cc|c} 1 & \frac{5}{2} & \frac{21}{2} \\ 0 & 1 & 5 \end{array} \right]$$

$$R_1 \leftarrow R_1 - \frac{5}{2} \times R_2$$

$$= \left[\begin{array}{cc|c} 1 & 0 & -2 \\ 0 & 1 & 5 \end{array} \right]$$

i. e.

$$x = -2$$

$$y = 5$$

Solution By Gauss jordan elimination method.

$$x = -2$$

$$y = 5$$

Week: 11

Topic: LU Factorization Method

Page: 21-26

Question: Solve the following system of linear equations using LU Factorization Method.

$$x + 2y + 4z = 3$$

$$3x + 8y + 14z = 13$$

$$2x + 6y + 13z = 4$$

Solution: Given system of equations is
$$\left. \begin{aligned} x + 2y + 4z &= 3 \\ 3x + 8y + 14z &= 13 \\ 2x + 6y + 13z &= 4 \end{aligned} \right\} \dots\dots (1)$$

Given system can be written as, $AX = b \dots\dots (2)$

where $A = \begin{pmatrix} 1 & 2 & 4 \\ 3 & 8 & 14 \\ 2 & 6 & 13 \end{pmatrix}$, $X = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$ and $b = \begin{pmatrix} 3 \\ 13 \\ 4 \end{pmatrix}$

Let $LU = A \dots\dots (3)$ where $L = \begin{pmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{pmatrix}$ and $U = \begin{pmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{pmatrix}$

Therefore $LUX = b \dots\dots (4)$

From (3) we get,
$$\begin{pmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{pmatrix} \begin{pmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{pmatrix} = \begin{pmatrix} 1 & 2 & 4 \\ 3 & 8 & 14 \\ 2 & 6 & 13 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} u_{11} & u_{12} & u_{13} \\ l_{21}u_{11} & l_{21}u_{12} + u_{22} & l_{21}u_{13} + u_{23} \\ l_{31}u_{11} & l_{31}u_{12} + l_{32}u_{22} & l_{31}u_{13} + l_{32}u_{23} + u_{33} \end{pmatrix} = \begin{pmatrix} 1 & 2 & 4 \\ 3 & 8 & 14 \\ 2 & 6 & 13 \end{pmatrix}$$

Now equating the corresponding components from both sides matrix, we get

$$u_{11} = 1, \quad u_{12} = 2, \quad u_{13} = 4$$

$$\begin{aligned} l_{21}u_{11} &= 3, & l_{21}u_{12} + u_{22} &= 8, & l_{21}u_{13} + u_{23} &= 14 \\ \therefore l_{21} &= 3, & \therefore u_{22} &= 8 - 6 = 2, & \therefore u_{23} &= 14 - 12 = 2 \end{aligned}$$

$$\begin{aligned} l_{31}u_{11} &= 2, & l_{31}u_{12} + l_{32}u_{22} &= 6, & l_{31}u_{13} + l_{32}u_{23} + u_{33} &= 13 \\ \therefore l_{31} &= 2, & \therefore l_{32} &= \frac{6-4}{2} = 1, & \therefore u_{33} &= 13 - 8 - 2 = 3 \end{aligned}$$

$$\text{Thus } L = \begin{pmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 2 & 1 & 1 \end{pmatrix} \quad \text{and} \quad U = \begin{pmatrix} 1 & 2 & 4 \\ 0 & 2 & 2 \\ 0 & 0 & 3 \end{pmatrix}.$$

$$\text{Let } UX = Y \quad \dots\dots (5) \text{ where } Y = \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix}$$

$$\text{So (4) implies } LY = b \Rightarrow \begin{pmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 2 & 1 & 1 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = \begin{pmatrix} 3 \\ 13 \\ 4 \end{pmatrix}$$

$$\Rightarrow \begin{cases} y_1 = 3 \\ 3y_1 + y_2 = 13 \\ 2y_1 + y_2 + y_3 = 4 \end{cases}$$

Now by forward substitution, we get

$$y_1 = 3, \quad y_2 = 13 - 9 = 4, \quad y_3 = 4 - 4 - 6 = -6 \quad \therefore Y = \begin{pmatrix} 3 \\ 4 \\ -6 \end{pmatrix}$$

$$\text{Now(5) becomes } \begin{pmatrix} 1 & 2 & 4 \\ 0 & 2 & 2 \\ 0 & 0 & 3 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 3 \\ 4 \\ -6 \end{pmatrix}$$

$$\Rightarrow \begin{cases} x + 2y + 4z = 3 \\ 2y + 2z = 4 \\ 3z = -6 \end{cases}$$

Now by backward substitution, we get

$$z = -2, \quad y = \frac{1}{2}(4 + 4) = 4, \quad x = 3 + 8 - 8 = 3 \quad \therefore X = \begin{pmatrix} 3 \\ 4 \\ -2 \end{pmatrix}$$

Hence the required solution of the system is $x = 3, \quad y = 4, \quad z = -2$.

Question: Solve the following system of linear equations using LU Factorization Method.

$$\begin{aligned}x + 5y + z &= 21 \\2x + y + 3z &= 20 \\3x + y + 4z &= 26\end{aligned}$$

Solution: Given system of equations is
$$\left. \begin{aligned}x + 5y + z &= 21 \\2x + y + 3z &= 20 \\3x + y + 4z &= 26\end{aligned} \right\} \dots\dots (1)$$

Given system can be written as, $AX = b \dots\dots (2)$

where $A = \begin{pmatrix} 1 & 5 & 1 \\ 2 & 1 & 3 \\ 3 & 1 & 4 \end{pmatrix}$, $X = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$ and $b = \begin{pmatrix} 20 \\ 21 \\ 26 \end{pmatrix}$

Let $LU = A \dots\dots (3)$ where $L = \begin{pmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{pmatrix}$ and $U = \begin{pmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{pmatrix}$

Therefore $LUX = b \dots\dots (4)$

From (3) we get,
$$\begin{pmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{pmatrix} \begin{pmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{pmatrix} = \begin{pmatrix} 1 & 5 & 1 \\ 2 & 1 & 3 \\ 3 & 1 & 4 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} u_{11} & u_{12} & u_{13} \\ l_{21}u_{11} & l_{21}u_{12} + u_{22} & l_{21}u_{13} + u_{23} \\ l_{31}u_{11} & l_{31}u_{12} + l_{32}u_{22} & l_{31}u_{13} + l_{32}u_{23} + u_{33} \end{pmatrix} = \begin{pmatrix} 1 & 5 & 1 \\ 2 & 1 & 3 \\ 3 & 1 & 4 \end{pmatrix}$$

Now equating the corresponding components from both sides matrix, we get

$$u_{11} = 1, \quad u_{12} = 5, \quad u_{13} = 1$$

$$\begin{aligned}l_{21}u_{11} &= 2, & l_{21}u_{12} + u_{22} &= 1, & l_{21}u_{13} + u_{23} &= 3 \\ \therefore l_{21} &= 2, & \therefore u_{22} &= 1 - 10 = -9, & \therefore u_{23} &= 3 - 2 = 1\end{aligned}$$

$$\begin{aligned}l_{31}u_{11} &= 3, & l_{31}u_{12} + l_{32}u_{22} &= 1, & l_{31}u_{13} + l_{32}u_{23} + u_{33} &= 4 \\ \therefore l_{31} &= 3, & \therefore l_{32} &= \frac{1 - 15}{-9} = \frac{14}{9}, & \therefore u_{33} &= 4 - 3 - \frac{14}{9} = -\frac{5}{9}\end{aligned}$$

Thus $L = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & \frac{14}{9} & 1 \end{pmatrix}$ and $U = \begin{pmatrix} 1 & 5 & 1 \\ 0 & -9 & 1 \\ 0 & 0 & -\frac{5}{9} \end{pmatrix}$. Let $UX = Y \dots\dots (5)$ where

$$Y = \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix}$$

So (4) implies $LY = b \Rightarrow \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & \frac{14}{9} & 1 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = \begin{pmatrix} 20 \\ 21 \\ 26 \end{pmatrix}$

$$\Rightarrow \begin{cases} y_1 = 20 \\ 2y_1 + y_2 = 21 \\ 3y_1 + \frac{14}{9}y_2 + y_3 = 26 \end{cases}$$

Now by forward substitution, we get

$$y_1 = 20, \quad y_2 = 21 - 40 = -19, \quad y_3 = 26 + \frac{14}{9} \cdot 19 - 60 = -\frac{40}{9} \quad \therefore Y = \begin{pmatrix} 20 \\ -19 \\ -\frac{40}{9} \end{pmatrix}$$

Now(5) becomes $\begin{pmatrix} 1 & 5 & 1 \\ 0 & -9 & 1 \\ 0 & 0 & -\frac{5}{9} \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 20 \\ -19 \\ -\frac{40}{9} \end{pmatrix}$

$$\Rightarrow \begin{cases} x + 5y + z = 20 \\ -9y + z = -19 \\ -\frac{5}{9}z = -\frac{40}{9} \end{cases}$$

Now by backward substitution, we get

$$z = 8, \quad y = -\frac{1}{9}(-19 - 8) = 3, \quad x = 20 - 15 - 8 = -3 \quad \therefore X = \begin{pmatrix} -3 \\ 3 \\ 8 \end{pmatrix}$$

Hence the required solution of the system is $x = -3$, $y = 3$, $z = 8$

Question: Solve the following system of linear equations using LU Factorization Method.

$$\begin{aligned}x + y + z &= 3 \\2x - y + 3z &= 16 \\3x + 2y - z &= -3\end{aligned}$$

Solution: Given system of equations is
$$\left. \begin{aligned}x + y + z &= 3 \\2x - y + 3z &= 16 \\3x + 2y - z &= -3\end{aligned} \right\} \dots\dots (1)$$

Given system can be written as, $AX = b \dots\dots (2)$

where $A = \begin{pmatrix} 1 & 1 & 1 \\ 2 & -1 & 3 \\ 3 & 2 & -1 \end{pmatrix}$, $X = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$ and $b = \begin{pmatrix} 3 \\ 16 \\ -3 \end{pmatrix}$

Let $LU = A \dots\dots (3)$ where $L = \begin{pmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{pmatrix}$ and $U = \begin{pmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{pmatrix}$

Therefore, $LUX = b \dots\dots (4)$

From (3) we get
$$\begin{pmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{pmatrix} \begin{pmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{pmatrix} = \begin{pmatrix} 1 & 1 & 1 \\ 2 & -1 & 3 \\ 3 & 2 & -1 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} u_{11} & u_{12} & u_{13} \\ l_{21}u_{11} & l_{21}u_{12} + u_{22} & l_{21}u_{13} + u_{23} \\ l_{31}u_{11} & l_{31}u_{12} + l_{32}u_{22} & l_{31}u_{13} + l_{32}u_{23} + u_{33} \end{pmatrix} = \begin{pmatrix} 1 & 1 & 1 \\ 2 & -1 & 3 \\ 3 & 1 & -1 \end{pmatrix}$$

Now equating the corresponding components from both sides matrix, we get

$$u_{11} = 1, \quad u_{12} = 1, \quad u_{13} = 1$$

$$\begin{aligned}l_{21}u_{11} &= 2, & l_{21}u_{12} + u_{22} &= -1, & l_{21}u_{13} + u_{23} &= 3 \\ \therefore l_{21} &= 2, & \therefore u_{22} &= -1 - 2 = -3, & \therefore u_{23} &= 3 - 2 = 1\end{aligned}$$

$$\begin{aligned}l_{31}u_{11} &= 3, & l_{31}u_{12} + l_{32}u_{22} &= 1, & l_{31}u_{13} + l_{32}u_{23} + u_{33} &= -1 \\ \therefore l_{31} &= 3, & \therefore l_{32} &= \frac{1-3}{-3} = \frac{2}{3}, & \therefore u_{33} &= -1 - 3 - \frac{2}{3} = -\frac{14}{3}\end{aligned}$$

Thus $L = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & \frac{2}{3} & 1 \end{pmatrix}$ and $U = \begin{pmatrix} 1 & 1 & 1 \\ 0 & -3 & 1 \\ 0 & 0 & -\frac{14}{3} \end{pmatrix}$. Let $UX = Y$ (5) where

$$Y = \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix}$$

So (4) implies $LY = b \Rightarrow \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & \frac{2}{3} & 1 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = \begin{pmatrix} 3 \\ 16 \\ -3 \end{pmatrix}$

$$\Rightarrow \begin{cases} y_1 = 3 \\ 2y_1 + y_2 = 16 \\ 3y_1 + \frac{2}{3}y_2 + y_3 = -3 \end{cases}$$

Now by forward substitution, we get

$$y_1 = 3, \quad y_2 = 16 - 6 = 10, \quad y_3 = -3 - \frac{2}{3} \times 10 - 3 \times 3 = -\frac{56}{3} \quad \therefore Y = \begin{pmatrix} 3 \\ 10 \\ -\frac{56}{3} \end{pmatrix}$$

Now (5) becomes $\begin{pmatrix} 1 & 1 & 1 \\ 0 & -3 & 1 \\ 0 & 0 & -\frac{14}{3} \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 3 \\ 10 \\ -\frac{56}{3} \end{pmatrix} \Rightarrow \begin{cases} x + y + z = 3 \\ -3y + z = 10 \\ -\frac{14}{3}z = -\frac{56}{3} \end{cases}$

Now by backward substitution, we get

$$z = 4, \quad y = \frac{1}{3}(-10 + 4) = -2, \quad x = 3 - (-2) - 4 = 1 \quad \therefore X = \begin{pmatrix} 1 \\ -2 \\ 4 \end{pmatrix}$$

Hence the required solution of the system is $x = 1, \quad y = -2, \quad z = 4$.

Exercise: Solve the following system of linear equations using LU Factorization Method.

$$\begin{array}{lll} x + 5y + z = 21 & 2x + 3y + z = 9 & x + y + z = 1 \\ (i) \quad 2x + y + 3z = 20 & (ii) \quad x + 2y + 3z = 6 & (iii) \quad 4x + 3y - z = 6 \\ 3x + y + 2z = 8 & 3x + y + 2z = 8 & 3x + 5y + 3z = 4 \end{array}$$

$$\begin{array}{lll} x + 2y + z = 3 & x + y + z = 3 & x + y + z = 1 \\ (iv) \quad 2x + 3y + z = 3 & (v) \quad x + 2y + 3z = 6 & (vi) \quad 3x + y - 3z = 5 \\ 3x - y - z = -6 & 2x + 3y + 5z = 10 & x - 2y - 5z = 10 \end{array}$$

Week: 12

Topic: Pivoting

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Pivoting (Basic Concept)

Pivoting is a technique used in numerical methods, particularly in solving systems of linear equations. It involves selecting a **pivot element** in the matrix and using it to simplify the system of equations. The pivoting process helps to improve the stability of the algorithm and avoid errors, such as dividing by very small numbers or zero.

There are two main types of pivoting:

1. **Partial Pivoting:** This involves selecting the largest element in the current column (in absolute value) to be the pivot. This helps reduce numerical errors by ensuring that the pivot is large, thus avoiding small numbers in the calculations.
2. **Complete Pivoting:** This is a more advanced form, where the largest element in the entire submatrix (both in the current row and column) is selected as the pivot. This provides even more stability, but it requires more computational work since it searches both rows and columns for the largest value.

In summary, **pivoting** ensures that the algorithm uses the most suitable elements in the matrix to prevent numerical issues and get accurate results when solving equations.

Week: 13+14

Topic: Forward and backward difference

Page: 27-32

Newton-Gregory Forward Difference formula:

$$y(x) = y_0 + u\Delta y_0 + \frac{u(u-1)}{2!}\Delta^2 y_0 + \frac{u(u-1)(u-2)}{3!}\Delta^3 y_0 + \dots + \frac{u(u-1)(u-2)\dots(u-n+1)}{n!}\Delta^n y_0$$
$$u = \frac{x - x_0}{h}$$

Newton-Gregory Backward Difference formula:

$$y(x) = y_n + u\nabla y_n + \frac{u(u+1)}{2!}\nabla^2 y_n + \frac{u(u+1)(u+2)}{3!}\nabla^3 y_n + \dots + \frac{u(u+1)(u+2)\dots(u+n-1)}{n!}\nabla^n y_n$$
$$u = \frac{x - x_n}{h}$$

Newton Divided Difference formula:

$$f(x) = f(x_0) + (x - x_0)f[x_0, x_1] + (x - x_0)(x - x_1)f[x_0, x_1, x_2] + \dots + (x - x_0)(x - x_1)(x - x_2)\dots(x - x_{n-1})f[x_0, x_1, x_2, \dots, x_n]$$

Lagrange's interpolation formula:

$$\begin{aligned} f(x) &= \frac{(x-x_1)(x-x_2)\cdots(x-x_n)}{(x_0-x_1)(x_0-x_2)\cdots(x_0-x_n)} f(x_0) + \frac{(x-x_0)(x-x_2)\cdots(x-x_n)}{(x_1-x_0)(x_1-x_2)\cdots(x_1-x_n)} f(x_1) \\ &+ \frac{(x-x_0)(x-x_1)(x-x_3)\cdots(x-x_n)}{(x_2-x_0)(x_2-x_1)(x_2-x_3)\cdots(x_2-x_n)} f(x_2) + \cdots + \frac{(x-x_0)(x-x_2)\cdots(x-x_{n-1})}{(x_n-x_0)(x_n-x_1)\cdots(x_n-x_{n-1})} f(x_n) \\ &= L_0(x) f(x_0) + L_1(x) f(x_1) + L_2(x) f(x_2) + \cdots + L_n(x) f(x_n) \end{aligned}$$

$$(x-a)(x-b) = x^2 - (a+b)x + ab$$

$$(x-a)(x-b)(x-c) = x^3 - (a+b+c)x^2 + (ab+bc+ca)x - abc$$

Newton-Gregory Forward Difference Formula

Question: The table below gives the values of $\tan x$ for $0.10 \leq x \leq 0.30$.

Estimate the value of (a) $\tan(0.12)$ (b) $\tan(0.26)$ (c) $\tan(0.40)$

x	0.10	0.15	0.20	0.25	0.30
$y = \tan x$	0.1003	0.1511	0.2027	0.2553	0.3093

Solution: We construct the difference table for the given data is as follows:

X	$f(x)$	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
0.10	0.1003	0.0508	0.0008	0.0002	0.0002
0.15	0.1511	0.0516	0.0010	0.0004	
0.20	0.2027	0.0526	0.0014		
0.25	0.2553	0.0540			
0.30	0.3093				

By Newton Forward Difference formula, we have

$$y(x) = y_0 + u\Delta y_0 + \frac{u(u-1)}{2!} \Delta^2 y_0 + \frac{u(u-1)(u-2)}{3!} \Delta^3 y_0 + \frac{u(u-1)(u-2)(u-3)}{4!} \Delta^4 y_0 + \cdots$$

(a) To find the value of $\tan(0.12)$, we have $x = 0.12$, $x_0 = 0.10$, $h = 0.05$

$$\text{Therefore } u = \frac{x - x_0}{h} = \frac{0.12 - 0.10}{0.05} = 0.4$$

$$\begin{aligned}\therefore y(0.12) &= \tan(0.12) = 0.1003 + 0.4 \times 0.0508 + \frac{0.4(0.4-1)}{2!} \times 0.0008 + \frac{0.4(0.4-1)(0.4-2)}{3!} \times 0.0002 \\ &\quad + \frac{0.4(0.4-1)(0.4-2)(0.4-3)}{4!} \times 0.0002 \\ &= 0.12052848\end{aligned}$$

(b) To find the value of $\tan(0.26)$, we have $x = 0.26$, $x_0 = 0.10$, $h = 0.05$.

Therefore $u = \frac{x - x_0}{h} = \frac{0.26 - 0.10}{0.05} = 3$

$$\begin{aligned}\therefore y(0.26) &= \tan(0.26) = 0.1003 + 3.2 \times 0.0508 + \frac{3.2(3.2-1)}{2!} \times 0.0008 + \frac{3.2(3.2-1)(3.2-2)}{3!} \times 0.0002 \\ &\quad + \frac{3.2(3.2-1)(3.2-2)(3.2-3)}{4!} \times 0.0002 \\ &= 0.26622768\end{aligned}$$

Question: The population of a town in the decennial census was as given below. Estimate the population for the year 1895 by using Forward difference formula.

Year x	1891	1901	1911	1921	1931
Population(in thousand) y	46	66	81	93	101

Solution: We construct the difference table for the given data is as follows:

X	$f(x)$	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
1891	46	20	-5	2	-3
1901	66	15	-3	-1	
1911	81	12	-4		
1921	93	8			
1931	101				

By Newton Forward Difference formula, we have

$$y(x) = y_0 + u\Delta y_0 + \frac{u(u-1)}{2!} \Delta^2 y_0 + \frac{u(u-1)(u-2)}{3!} \Delta^3 y_0 + \frac{u(u-1)(u-2)(u-3)}{4!} \Delta^4 y_0 + \dots$$

(a) To find the value of $y(1895)$, we have $x = 1895$, $x_0 = 1891$, $h = 10$

Therefore $u = \frac{x - x_0}{h} = \frac{1895 - 1891}{10} = 0.4$

$$\begin{aligned}y(1895) &= 46 + 0.4 \times 20 + \frac{0.4(0.4-1)}{2!} \times (-5) + \frac{0.4(0.4-1)(0.4-2)}{3!} \times 2 + \frac{0.4(0.4-1)(0.4-2)(0.4-3)}{4!} \times (-3) \\ &= 54.853\end{aligned}$$

The population for the year 1895 is 54.853 thousands approximately.

Exercise-1: From the following table of premium for policies maturing at different ages, estimate the premium for policies maturing at age 46 and 63.

Age x	45	50	55	60	65
Premium y	114.84	96.16	83.32	74.48	68.48

Exercise-2: Estimate using forward difference method the increase in the population during the year from 1946 to 1948.

Year	1911	1921	1931	1941	1951	1961
Population(thousands)	12	15	20	27	39	52

Question-1: Marks obtain by students in an examination are given below. Estimate the number of students who obtained less than (b) 90 marks.

Marks obtain (less than)	20	40	60	80	100
No. of students	41	103	168	218	235

Solution: We construct a difference table of the given data is as follows:

Marks obtain (less than) X	No. of students $f(x)$	$\nabla f(x)$	$\nabla^2 f(x)$	$\nabla^3 f(x)$	$\nabla^4 f(x)$
20	41				
40	103	62			
60	168	65	3		
80	218	50	-15	-18	
100	235	17	-33	-18	0

(b) From the table we shall find $f(90)$, the number of students who get less than 90 marks.

In this case, we use backward difference formula. -

$$\text{Here } u = \frac{x-x_n}{h} = \frac{90-100}{20} = -0.5$$

$$f(90) = 235 + (-0.5) \times 17 + \frac{-0.5(-0.5+1)}{2!} \times (-33) + \frac{-0.5(-0.5+1)(-0.5+2)}{3!} \times (-18)$$

$$= 231.75$$

Hence the number of students who get less than 90 marks is 231.75.

Question-2: The following data satisfying $y = f(x)$. Find the Backward difference formula.

x	1	2	3	4	5	6
$f(x)$	0	15	48	105	192	315

Solution: We construct a backward difference table of the given data is as follows:

X	$f(x)$	$\nabla f(x)$	$\nabla^2 f(x)$	$\nabla^3 f(x)$	$\nabla^4 f(x)$
1	0				
2	15	15			
3	48	33	18		
4	105	57	24	6	
5	192	87	30	6	0
6	315	123	36	6	0

By Newton Backward Difference formula, we have

$$f(x) = f(x_n) + u \nabla f(x_n) + \frac{u(u+1)}{2!} \nabla^2 f(x_n) + \frac{u(u+1)(u+2)}{3!} \nabla^3 f(x_n) + \dots$$

$$\text{Here } u = \frac{x - x_n}{h} = \frac{x - 6}{1} = x - 6$$

$$\begin{aligned} f(x) &= 315 + (x-6)123 + \frac{(x-6)(x-5)}{2!} \times 36 + \frac{(x-6)(x-5)(x-4)}{3!} \times 6 \\ &= 315 + 123x - 738 + 18(x^2 - 11x + 30) + (x^2 - 11x + 30)(x-4) \\ &= 315 + 123x - 738 + 18x^2 - 198x + 540 + x^3 - 4x^2 - 11x^2 + 44x + 30x - 120 \\ &= x^3 + 3x^2 - x - 3 \end{aligned}$$

Question-3: The following data satisfying $y = f(x)$. Find the Backward difference formula.

x	1	2	3	4	5	6
$f(x)$	2	6	20	50	102	182

Solution: We construct a backward difference table of the given data is as follows:

X	$f(x)$	$\nabla f(x)$	$\nabla^2 f(x)$	$\nabla^3 f(x)$	$\nabla^4 f(x)$
1	2				

2	6	4			
3	20	14	10		
4	50	30	16	6	
5	102	52	22	6	0
6	182	80	28	6	0

By Newton Backward Difference formula, we have

$$f(x) = f(x_n) + u\nabla f(x_n) + \frac{u(u+1)}{2!} \nabla^2 f(x_n) + \frac{u(u+1)(u+2)}{3!} \nabla^3 f(x_n) + \dots$$

$$\text{Here } u = \frac{x - x_n}{h} = \frac{x - 6}{1} = x - 6$$

$$\begin{aligned} f(x) &= 182 + (x-6)80 + \frac{(x-6)(x-5)}{2!} 28 + \frac{(x-6)(x-5)(x-4)}{3!} 6 \\ &= 182 + 80x - 480 + 14(x^2 - 11x + 30) + x^3 - 15x^2 + 74x - 120 \\ &= x^3 - x^2 + 2 \end{aligned}$$

Exercise-1: Population was recorded as follows in a village. Estimate the population for the 1985 using Newton-Gregory Backward difference formula.

x	1941	1951	1961	1971	1981	1991
y	2500	2800	3200	3700	4350	5225

Exercise-2: Marks obtain by students in an examination are given below. Estimate the number of students who obtained less than (a) 45 marks (b) 78 marks.

Marks obtain (less than)	40	50	60	70	80
No. of students	31	73	124	159	190

Week: 15

Topic: Divided difference

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Question-1: Given the following data satisfying $y = f(x)$. Find the Divided difference formula.

x	-1	0	3	6	7
y	3	-6	39	822	1611

Solution: Since the interval is unequal, so we construct the following divided difference table:

X	$f(x)$	$\Delta f(x)$	$\Delta^2 f(x)$	$\Delta^3 f(x)$	$\Delta^4 f(x)$
-1	3	$\frac{-6-3}{0+1} = -9$			
0	-6	$\frac{39+6}{3-0} = 15$	$\frac{15+9}{3+1} = 6$		
3	39	$\frac{822-39}{6-3} = 261$	$\frac{261-15}{6-0} = 41$	$\frac{41-6}{6+1} = 5$	
6	822	$\frac{1611-822}{7-6} = 789$	$\frac{789-261}{7-3} = 132$	$\frac{132-41}{7-0} = 13$	$\frac{13-5}{7+1} = 1$
7	1611				

By Newton divided difference formula, we have

$$f(x) = f(x_0) + (x-x_0)\Delta f(x_0) + (x-x_0)(x-x_1)\Delta^2 f(x_0) + (x-x_0)(x-x_1)(x-x_2)\Delta^3 f(x_0) + (x-x_0)(x-x_1)(x-x_2)(x-x_3)\Delta^4 f(x_0)$$

$$= 3 + (x+1)(-9) + (x+1)(x-0) \cdot 6 + (x+1)(x-0)(x-3) \cdot 5 + (x+1)(x-0)(x-3)(x-6) \cdot 1$$

$$= 3 - 9x - 9 + 6(x^2 + x) + 5x(x^2 - 2x - 3) + (x^2 + x)(x^2 - 9x + 18)$$

$$= 3 - 9x - 9 + 6x^2 + 6x + 5x^3 - 10x^2 - 15x + x^4 - 9x^3 + 18x^2 + x^3 - 9x^2 + 18x$$

$$= x^4 - 3x^3 + 5x^2 - 6$$

Question-2: Given the following data satisfying $y = f(x)$. Find the Divided difference formula.

x	0	2	3	4	7	9
y	4	26	58	112	466	922

Solution: Since the interval is unequal, so we construct the following divided difference table:

X	$f(x)$	$\Delta f(x)$	$\Delta^2 f(x)$	$\Delta^3 f(x)$	$\Delta^4 f(x)$
-----	--------	---------------	-----------------	-----------------	-----------------

0	4	$\frac{26-4}{2-0} = 11$			
2	26	$\frac{58-26}{3-2} = 32$	$\frac{32-11}{3-0} = 7$		
3	58	$\frac{112-58}{4-3} = 54$	$\frac{54-32}{4-2} = 11$	$\frac{11-7}{4-0} = 1$	
4	112	$\frac{466-122}{7-4} = 118$	$\frac{466-122}{7-3} = 16$	$\frac{16-11}{7-2} = 1$	$\frac{1-1}{7-1} = 0$
7	446			$\frac{22-16}{9-3} = 1$	$\frac{1-1}{9-2} = 0$
9	912	$\frac{922-466}{9-7} = 228$	$\frac{228-118}{9-4} = 22$		

By Newton divided difference formula, we have

$$f(x) = f(x_0) + (x-x_0)\Delta f(x_0) + (x-x_0)(x-x_1)\Delta^2 f(x_0) + (x-x_0)(x-x_1)(x-x_2)\Delta^3 f(x_0) + (x-x_0)(x-x_1)(x-x_2)(x-x_3)\Delta^4 f(x_0)$$

$$= 4 + (x-0) \times 11 + (x-0)(x-2) \times 7 + (x-0)(x-2)(x-3) \times 1 + (x-0)(x-2)(x-3)(x-4) \times 0$$

$$= 4 + 11x + 7(x^2 - 2x) + x(x^2 - 5x + 6)$$

$$= 4 + 11x + 7x^2 - 14x + x^3 - 5x^2 + 6x$$

$$= x^3 + 2x^2 + 3x + 4$$

Question-3: Given the following data satisfying $y = f(x)$. Find the Divided difference formula.

x	1	3	4	6	7
y	-3	9	30	132	225

Solution: Since the interval is unequal, so we construct the following divided difference table:

X	$f(x)$	$\Delta f(x)$	$\Delta^2 f(x)$	$\Delta^3 f(x)$	$\Delta^4 f(x)$
---	--------	---------------	-----------------	-----------------	-----------------

1	-3	$\frac{9+3}{3-1} = 6$			
3	9	$\frac{30-9}{4-3} = 21$	$\frac{21-6}{4-1} = 5$		
4	30	$\frac{132-30}{6-4} = 51$	$\frac{51-21}{6-3} = 10$	$\frac{10-5}{6-1} = 1$	
6	132	$\frac{225-132}{7-6} = 93$	$\frac{93-51}{7-4} = 14$	$\frac{14-10}{7-3} = 1$	$\frac{1-1}{7-1} = 0$
7	225				

By Newton divided difference formula, we have

$$\begin{aligned}
 f(x) &= f(x_0) + (x-x_0)\Delta f(x_0) + (x-x_0)(x-x_1)\Delta^2 f(x_0) + (x-x_0)(x-x_1)(x-x_2)\Delta^3 f(x_0) \\
 &\quad + (x-x_0)(x-x_1)(x-x_2)(x-x_3)\Delta^4 f(x_0) \\
 &= -3 + (x-1) \times 6 + (x-1)(x-3) \times 5 + (x-1)(x-3)(x-4) \times 1 + (x-1)(x-3)(x-4)(x-6) \times 0 \\
 &= -3 + 6x - 6 + 5(x^2 - 4x + 3) + (x-1)(x^2 - 7x + 12) + 0 \\
 &= -3 + 6x - 6 + 5x^2 - 20x + 15 + x^3 - 7x^2 + 12x - x^2 + 7x - 12 \\
 &= x^3 - 3x^2 + 5x - 6
 \end{aligned}$$

Exercise-1: Given the following data satisfying $y = f(x)$. Find the Divided difference formula.

x	4	5	7	10	11	13
y	48	100	294	900	1210	2028

Exercise-2: Given the following data satisfying $y = f(x)$. Find the Divided difference formula.

x	-1	0	1	4	5
y	3	8	11	68	123

Week: 16+17

Topic: Lagrange Polynomial

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Answer: (1) $f(x) = x^3 - x^2$ (2) $f(x) = x^3 - x^2 + 3x + 8$

Question-1: Given the following data satisfying-. Find the Lagrange Polynomial.

x	-1	2	3	5
y	35	32	15	-7

Solution: Here, $x_0 = -1$, $x_1 = 2$, $x_2 = 3$, $x_3 = 5$ and $y_0 = 35$, $y_1 = 32$, $y_2 = 15$, $y_3 = -7$

By Lagrange interpolation formula, we have

$$f(x) = L_0(x)y_0 + L_1(x)y_1 + L_2(x)y_2 + \dots + L_n(x)y_n \quad \dots (1)$$

Where

$$L_0(x) = \frac{(x-x_1)(x-x_2)(x-x_3)}{(x_0-x_1)(x_0-x_2)(x_0-x_3)} = \frac{(x-2)(x-3)(x-5)}{(-3)(-4)(-6)} = \frac{1}{-72}(x^3 - 10x^2 + 31x - 30)$$

$$L_1(x) = \frac{(x-x_0)(x-x_2)(x-x_3)}{(x_1-x_0)(x_1-x_2)(x_1-x_3)} = \frac{(x+1)(x-3)(x-5)}{3 \times (-1) \times (-3)} = \frac{1}{9}(x^3 - 7x^2 + 7x + 15)$$

$$L_2(x) = \frac{(x-x_0)(x-x_1)(x-x_3)}{(x_2-x_0)(x_2-x_1)(x_2-x_3)} = \frac{(x+1)(x-2)(x-5)}{4 \times 1 \times (-2)} = \frac{1}{-8}(x^3 - 6x^2 + 3x + 10)$$

$$\text{and } L_3(x) = \frac{(x-x_0)(x-x_1)(x-x_2)}{(x_3-x_0)(x_3-x_1)(x_3-x_2)} = \frac{(x+1)(x-2)(x-3)}{6 \times 3 \times 2} = \frac{1}{36}(x^3 - 4x^2 + x + 6)$$

From (1), we get

$$\begin{aligned} y(x) &= -\frac{1}{72}(x^3 - 10x^2 + 31x - 30) \times 35 + \frac{1}{9}(x^3 - 7x^2 + 7x + 15) \times 32 \\ &\quad - \frac{1}{8}(x^3 - 6x^2 + 3x + 10) \times 15 + \frac{1}{36}(x^3 - 4x^2 + x + 6) \times (-7) \\ &= -\frac{35}{72}x^3 + \frac{350}{72}x^2 - \frac{1085}{72}x + \frac{1050}{72} + \frac{32}{9}x^3 - \frac{224}{9}x^2 + \frac{224}{9}x + \frac{480}{9} \\ &\quad - \frac{15}{8}x^3 + \frac{90}{8}x^2 - \frac{45}{8}x - \frac{150}{8} - \frac{7}{36}x^3 + \frac{28}{36}x^2 - \frac{7}{36}x - \frac{42}{36} \\ &= \left(-\frac{35}{72} + \frac{32}{9} - \frac{15}{8} - \frac{7}{36}\right)x^3 + \left(\frac{350}{72} - \frac{224}{9} + \frac{90}{8} + \frac{28}{36}\right)x^2 + \left(-\frac{1085}{72} + \frac{224}{9} - \frac{45}{8} - \frac{7}{36}\right)x \\ &\quad + \left(\frac{1050}{72} + \frac{480}{9} - \frac{150}{8} - \frac{42}{36}\right) \\ &= x^3 - 8x^2 + 4x + 48 \end{aligned}$$

Question-2: Given the following data satisfying $y = f(x)$. Find the Lagrange Polynomial.

x	0	1	2	5
y	24	18	8	14

Solution: Here, $x_0 = 0$, $x_1 = 1$, $x_2 = 2$, $x_3 = 5$ and $y_0 = 24$, $y_1 = 18$, $y_2 = 8$, $y_3 = 14$

By Lagrange interpolation formula, we have

$$f(x) = L_0(x)y_0 + L_1(x)y_1 + L_2(x)y_2 + \dots + L_n(x)y_n \quad \dots (1)$$

$$\text{Where } L_0(x) = \frac{(x-x_1)(x-x_2)(x-x_3)}{(x_0-x_1)(x_0-x_2)(x_0-x_3)} = \frac{(x-1)(x-2)(x-5)}{(-1)(-2)(-5)} = \frac{1}{-10}(x^3 - 8x^2 + 17x - 10)$$

$$L_1(x) = \frac{(x-x_0)(x-x_2)(x-x_3)}{(x_1-x_0)(x_1-x_2)(x_1-x_3)} = \frac{(x-0)(x-2)(x-5)}{1(-1)(-4)} = \frac{1}{4}(x^3 - 7x^2 + 10x)$$

$$L_2(x) = \frac{(x-x_0)(x-x_1)(x-x_3)}{(x_2-x_0)(x_2-x_1)(x_2-x_3)} = \frac{(x-0)(x-1)(x-5)}{2 \times 1 \times (-3)} = \frac{1}{-6}(x^3 - 6x^2 + 5x)$$

$$\text{and } L_3(x) = \frac{(x-x_0)(x-x_1)(x-x_2)}{(x_3-x_0)(x_3-x_1)(x_3-x_2)} = \frac{(x-0)(x-1)(x-2)}{5 \times 4 \times 3} = \frac{1}{60}(x^3 - 3x^2 + 2x)$$

From (1), we get

$$y(x) = -\frac{1}{10}(x^3 - 8x^2 + 17x - 10) \times (24) + \frac{1}{4}(x^3 - 7x^2 + 10x) \times 18 - \frac{1}{6}(x^3 - 6x^2 + 5x) \times 8 + \frac{1}{60}(x^3 - 3x^2 + 2x) \times 14$$

$$= -\frac{12}{5}x^3 + \frac{96}{5}x^2 - \frac{204}{5}x + 24 + \frac{9}{2}x^3 - \frac{63}{2}x^2 + 45x - \frac{4}{3}x^3 + 8x^2 - \frac{20}{3}x + \frac{7}{30}x^3 - \frac{7}{10}x^2 + \frac{7}{15}x$$

$$= \left(-\frac{12}{5} + \frac{9}{2} - \frac{4}{3} + \frac{7}{30}\right)x^3 + \left(\frac{96}{5} - \frac{63}{2} + 8 - \frac{7}{10}\right)x^2 + \left(-\frac{204}{5} + 45 - \frac{20}{3} + \frac{7}{15}\right)x + 24$$

$$= x^3 - 5x^2 - 2x + 24$$

Question-3: Given the following data satisfying $y = f(x)$. Find the Lagrange Polynomial.

x	0	1	3	4
y	-12	0	12	24

Solution: Here, $x_0 = 0$, $x_1 = 1$, $x_2 = 3$, $x_3 = 4$ and $y_0 = -12$, $y_1 = 0$, $y_2 = 12$, $y_3 = 24$

By Lagrange interpolation formula, we have

$$f(x) = L_0(x)y_0 + L_1(x)y_1 + L_2(x)y_2 + \dots + L_n(x)y_n \quad \dots (1)$$

$$\text{Where } L_0(x) = \frac{(x-x_1)(x-x_2)(x-x_3)}{(x_0-x_1)(x_0-x_2)(x_0-x_3)} = \frac{(x-1)(x-3)(x-4)}{(-1)(-3)(-4)} = \frac{1}{-12}(x^3 - 8x^2 + 19x - 12)$$

$$L_1(x) = \frac{(x-x_0)(x-x_2)(x-x_3)}{(x_1-x_0)(x_1-x_2)(x_1-x_3)} = \frac{(x-0)(x-3)(x-4)}{1(-2)(-3)} = \frac{1}{6}(x^3 - 7x^2 + 12x)$$

$$L_2(x) = \frac{(x-x_0)(x-x_1)(x-x_3)}{(x_2-x_0)(x_2-x_1)(x_2-x_3)} = \frac{(x-0)(x-1)(x-4)}{3 \times 2 \times (-1)} = \frac{1}{-6}(x^3 - 5x^2 + 4x)$$

$$\text{and } L_3(x) = \frac{(x-x_0)(x-x_1)(x-x_2)}{(x_3-x_0)(x_3-x_1)(x_3-x_2)} = \frac{(x-0)(x-1)(x-3)}{4 \times 3 \times 1} = \frac{1}{12}(x^3 - 4x^2 + 3x)$$

From (1), we get

$$\begin{aligned} y(x) &= \frac{1}{-12}(x^3 - 8x^2 + 19x - 12) \times (-12) + 0 - \frac{1}{6}(x^3 - 5x^2 + 4x) \times 12 + \frac{1}{12}(x^3 - 4x^2 + 3x) \times 24 \\ &= x^3 - 8x^2 + 19x - 12 - 2x^3 + 10x^2 - 8x - 12 + 2x^3 - 8x^2 + 6x \\ &= x^3 - 6x^2 + 17x - 12 \end{aligned}$$

Exercise-1: Given the following data satisfying $y = f(x)$. Find the Lagrange Polynomial.

x	1	3	4	6
y	-10	36	110	420

Exercise-2: Given the following data satisfying $y = f(x)$. Find the Lagrange Polynomial.

x	1	3	5	6
y	18	-8	30	88

Exercise-3: Given the following data satisfying $y = f(x)$. Find the Lagrange Polynomial.

x	1	3	4	6
y	-3	9	30	132

Answer:

$$1. f(x) = 2x^3 + x^2 - 7x - 6 \quad 2. f(x) = x^3 - x^2 - 22x + 40 \quad 3. f(x) = x^3 - 3x^2 + 5x - 6$$